SQL Practice Assignment 2: Advanced Operators & Logic

Topic: AND, OR, BETWEEN, IN, IS NULL, NOT, and Logic Precedence

Role: Junior Data Analyst at "TechTrendz Retail"

Part 1: Data Preparation (Update Your Dataset)

Context: The product team has added two new experimental items to the system. One is a prototype with no ratings yet, and the other is a generic item with no specific category. You need to add these to your existing products table to perform the analysis.

Task: Write the SQL INSERT statements to add these two rows:

ID	Product Name	Category	Price	Stock	Rating
16	Smart Glasses Prototype	Electronics	1500.00	5	*NULL* (Leave empty)
17	Mystery Grab Bag	*NULL* (Leave empty)	15.00	50	3.5

Tip: When inserting a NULL value, you simply type NULL (without quotes) or omit the column if your table allows it.

```
USE techtrendz_retail;

-- Insert a new product with no rating yet (rating is NULL)

INSERT INTO products VALUES

(16,'Smart Glasses Prototype','Electronics',1500.00,5,NULL);

-- Insert a product with no category (category is NULL)

INSERT INTO products VALUES

(17,'Mystery Grab Bag',NULL,15.00,50,3.5);
```

```sql
USE techtrendz_retail;
-- Insert a new product with no rating yet (rating is NULL)
INSERT INTO products VALUES
(16,'Smart Glasses Prototype','Electronics',1500.00,5,NULL);

-- Insert a product with no category (category is NULL)
INSERT INTO products VALUES
(17,'Mystery Grab Bag',NULL,15.00,50,3.5);
```

Output

Action Output

#	Time	Action	Message
1	20:52:38	USE techtrendz_retail	0 row(s) affected
2	20:52:43	INSERT INTO products VALUES (16,'Smart Glasses Prototype','Electronics',1500.00,5,NULL)	1 row(s) affected
3	20:52:43	INSERT INTO products VALUES (17,'Mystery Grab Bag',NULL,15.00,50,3.5)	1 row(s) affected

Part 2: Advanced Analysis Questions

Write SQL queries to solve the following complex business requests.

A. Multiple Conditions (AND / OR)

1. **Premium Electronics:** We need to find our high-end tech. Select all columns for products where the Category is **'Electronics'** AND the Price is **greater than $300**.

```sql
-- Find Electronics products that cost more than $300
SELECT *
FROM products
WHERE category = 'Electronics'
AND price > 300;
```

```sql
-- Find Electronics products that cost more than $300
SELECT *
FROM products
WHERE category = 'Electronics'
AND price > 300;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	product_name	category	price	stock_quantity	rating
3	Gaming Laptop Pro	Electronics	1200.00	10	4.9
8	4K Smart TV 55 inch	Electronics	600.00	15	4.7
16	Smart Glasses Prototype	Electronics	1500.00	5	NULL
NULL	NULL	NULL	NULL	NULL	NULL

2. **Lifestyle Products:** The marketing team is running a joint promo for clothing and fitness gear. Select the product_name and category for items where the Category is either **'Apparel'** OR **'Fitness'**.

   -- Select products that belong to Apparel or Fitness category
   SELECT product_name, category
   FROM products
   WHERE category = 'Apparel'
   OR category = 'Fitness';

```sql
-- Select products that belong to Apparel or Fitness category
SELECT product_name, category
FROM products
WHERE category = 'Apparel'
OR category = 'Fitness';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	category
Running Shoes - Size 10	Apparel
Yoga Mat	Fitness
Cotton T-Shirt	Apparel

B. Ranges and Lists (BETWEEN / IN)

3. **Mid-Range Gift Guide:** We want to feature items that are affordable but not "cheap". Select product_name and price for items where the price is **BETWEEN $40 AND $100** (inclusive).

-- Find products with price between $40 and $100 (inclusive)
SELECT product_name, price
FROM products
WHERE price BETWEEN 40 AND 100;

```
-- Find products with price between $40 and $100 (inclusive)
SELECT product_name, price
FROM products
WHERE price BETWEEN 40 AND 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	price
Bluetooth Speaker Mini	45.99
Running Shoes - Size 10	89.99
Leather Wallet	45.00
Coffee Maker	75.50
Gaming Mouse	50.00

4. **Home Goods Selection:** We need a report on items for the home. Select all columns for products where the Category is **IN** the list: **'Furniture', 'Kitchen', 'Appliances'**.

-- Select products that belong to Furniture, Kitchen, or Appliances category

SELECT *

FROM products

WHERE category IN ('Furniture','Kitchen','Appliances');

```
-- Select products that belong to Furniture, Kitchen, or Appliances category
SELECT *
FROM products
WHERE category IN ('Furniture','Kitchen','Appliances');
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	rating
2	Ergonomic Office Chair	Furniture	150.00	20	4.5
10	Coffee Maker	Appliances	75.50	40	4.4
14	Desk Lamp	Furniture	35.00	0	4.0
15	Ceramic Mug Set	Kitchen	25.00	80	4.2
NULL	NULL	NULL	NULL	NULL	NULL

C. Handling Missing Data (IS NULL / IS NOT NULL)

5. **Unrated Products:** Customer service needs to identify items that haven't been rated yet to ask for user feedback. Select the product_name for items where the Rating **IS NULL**.

   -- Find products where rating is missing (NULL)
   SELECT product_name
   FROM products
   WHERE rating IS NULL;

```
-- Find products where rating is missing (NULL)
SELECT product_name
FROM products
WHERE rating IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name
Smart Glasses Prototype

6. **Uncategorized Audit:** The inventory manager needs to fix data errors. Select all columns for products where the Category **IS NULL**.

   -- Find products where category is missing (NULL)

   SELECT *

   FROM products

   WHERE category IS NULL;

```
-- Find products where category is missing (NULL)
SELECT *
FROM products
WHERE category IS NULL;
```

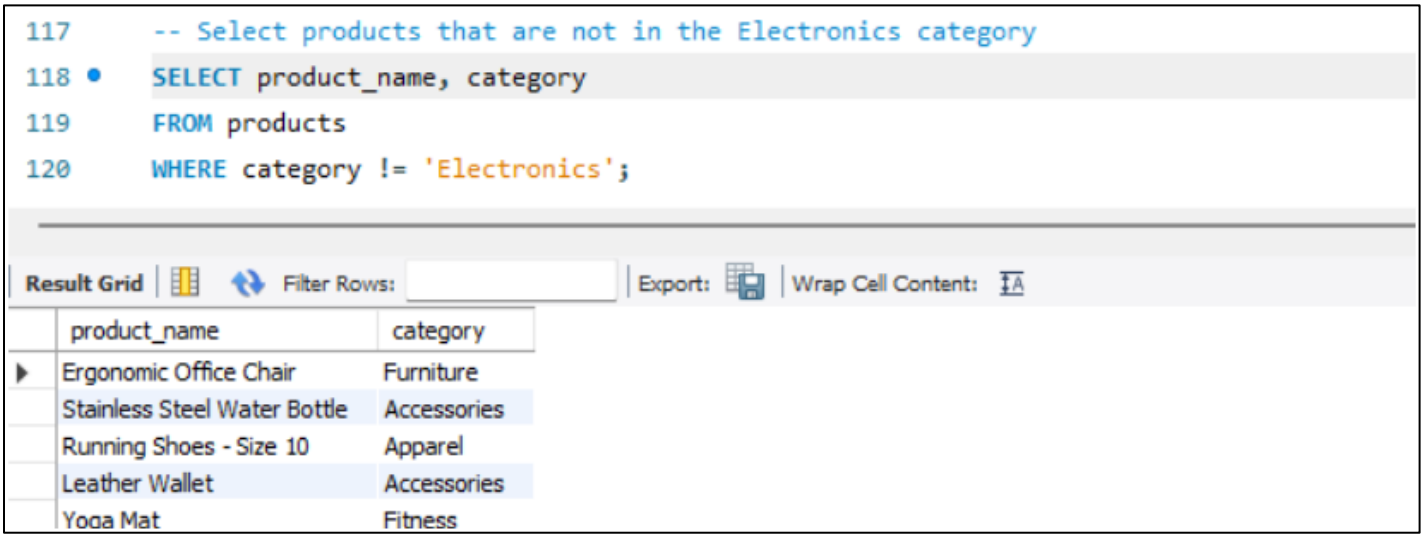
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

product_id	product_name	category	price	stock_quantity	rating
17	Mystery Grab Bag	NULL	15.00	50	3.5
NULL	NULL	NULL	NULL	NULL	NULL

D. Exclusion (NOT)

7. **Non-Electronic Items:** We want to analyze our non-tech inventory. Select product_name and category for all items where the Category is **NOT 'Electronics'**.

```
-- Select products that are not in the Electronics category
SELECT product_name, category
FROM products
WHERE category != 'Electronics';
```



E. Logic Priority (Parentheses & Precedence)

This is the most critical section for interviews. Pay attention to how you group your logic!

8. **Affordable Accessories & Apparel:** We want to find cheap items (under $50) that are either 'Accessories' OR 'Apparel'.
   - *Requirement:* Select product_name, category, and price.
   - *Logic:* The category must be 'Accessories' OR 'Apparel', **AND** the price must be less than $50.
   - *Hint:* Use parentheses () to group the OR condition correctly.

```
-- Find cheap items under $50 that belong to Accessories OR Apparel
-- Parentheses ensure OR condition is evaluated first
SELECT product_name, category, price
FROM products
WHERE (category = 'Accessories' OR category = 'Apparel')
AND price < 50;
```

```sql
-- Find cheap items under $50 that belong to Accessories OR Apparel
-- Parentheses ensure OR condition is evaluated first
SELECT product_name, category, price
FROM products
WHERE (category = 'Accessories' OR category = 'Apparel')
AND price < 50;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	category	price
Stainless Steel Water Bottle	Accessories	25.00
Leather Wallet	Accessories	45.00
Cotton T-Shirt	Apparel	15.00

9. **The "High Stock" Exception:**
   - Find products that are NOT 'Electronics', but still have a stock quantity greater than 80.

     -- Find products that are NOT Electronics but have stock greater than 80
     SELECT * FROM products
     WHERE category != 'Electronics'
     AND stock_quantity > 80;

```sql
-- Find products that are NOT Electronics but have stock greater than 80
SELECT *FROM products
WHERE category != 'Electronics'
AND stock_quantity > 80;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	rating
4	Stainless Steel Water Bottle	Accessories	25.00	100	4.2
9	Yoga Mat	Fitness	20.00	200	4.3
12	Cotton T-Shirt	Apparel	15.00	300	3.9
NULL	NULL	NULL	NULL	NULL	NULL